10 (a) State Wien's displacement law.

.....

..... [1]

(b) Fig. 10.1 shows the wavelength distributions of electromagnetic radiation emitted by two stars A and B.

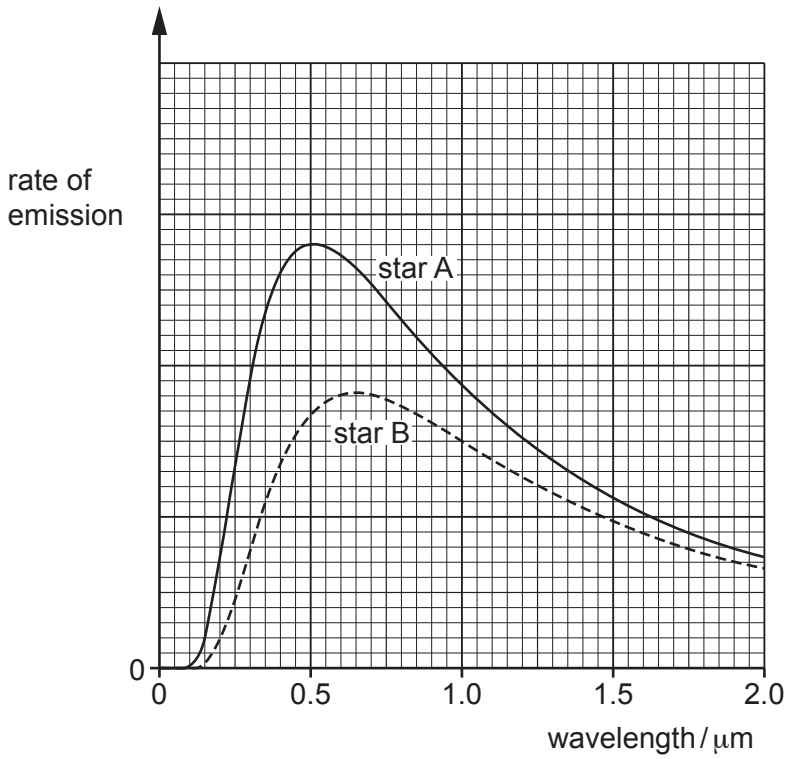


Fig. 10.1

The surface temperature of star A is known to be 5800 K.

(i) Determine the surface temperature of star B.

surface temperature = ... K [2]

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(ii) Star B appears less bright than star A when viewed from the Earth.

Use Fig. 10.1 to suggest, with a reason, how else the physical appearance of star B compares with that of star A.

.....

.....

..... [2]

(c) The lines in Fig. 10.1 have been corrected for redshift.

(i) State what is meant by redshift.

.....

.....

..... [2]

(ii) Explain how cosmologists are able to determine that light from a distant star has undergone redshift.

.....

.....

..... [2]

[Total: 9]